

Robotics Semantic Search Reference

1. Robot Perception

Robot perception uses sensors such as cameras, LiDAR, radar, and ultrasonic sensors to understand objects, distances, motion, and the surrounding environment.

2. LiDAR

LiDAR uses laser pulses to measure distances and create detailed representations of the environment. It is used for mapping, localization, navigation, and obstacle detection.

3. Camera Sensors

Cameras capture visual information that robots can process for object detection, recognition, tracking, and navigation.

4. Radar

Radar uses radio waves to detect objects and estimate properties such as distance and velocity. It can be useful when visibility is limited.

5. Ultrasonic Sensors

Ultrasonic sensors use high-frequency sound waves to estimate the distance to nearby objects and are commonly used for short-range obstacle detection.

6. Obstacle Avoidance

Obstacle avoidance combines sensor information with planning and control. A robot detects obstacles, estimates their position, and selects a collision-free path.

7. Sensor Fusion

Sensor fusion combines information from multiple sensors to improve the reliability and completeness of robot perception.

8. SLAM

Simultaneous Localization and Mapping, or SLAM, allows a robot to estimate its position while building a map of an unknown environment.

9. Path Planning

Path planning determines a route from a robot's current position to a target while considering obstacles and movement constraints. A* and RRT are examples of planning algorithms.

10. Autonomous Navigation

Autonomous navigation combines perception, localization, planning, and control so a robot can use sensor information to understand its environment and adjust its movement.